

# Aluminium powers next-generation battery design



EQONIC Group confirms aluminium as the chemistry behind its breakthrough battery technology  
(Credit: <https://batterytechassociation.org>)

EQONIC Group has announced an aluminium-based battery chemistry designed to eliminate lithium, sodium, and rare-earth materials, advancing a potential alternative for next-generation energy storage. Developed by the UK-based company, the architecture is inherently non-flammable and targets material costs at about 30% of conventional lithium-metal batteries, with a projected manufacturing cost of £50/kWh at scale. The technology could serve electric mobility, renewable-energy storage, and industrial applications while reducing exposure to critical-material supply constraints. EQONIC is now progressing the battery towards industrial-scale validation to assess manufacturability, reliability, and performance before commercial deployment.

# Single-electron memory advances AI chips

Researchers at Fudan University have developed a two-dimensional flash memory that stores one bit using a single electron, reducing AI chip power consumption. The device uses graphene to prevent electron leakage and amplify the stored charge signal to about 0.5V. Its architecture demonstrates voltage quantisation and self-limiting programming, opening possibilities for higher-density, multi-state memory. Developed in Shanghai, the technology could support energy-efficient AI applications, including language models on mobile devices. The team says the architecture is progressing toward commercialisation, and is likely to reach the market within three to five years.

The 2D flash chip features a layer of graphene that prevents electrons from escaping (Credit: [www.scmp.com](http://www.scmp.com))

